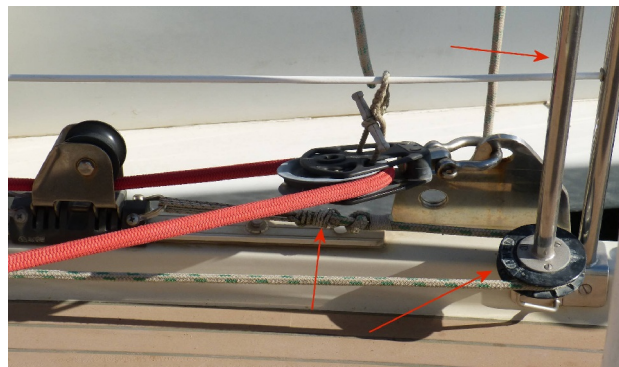


**Instruction manual
Replacement of toe pulleys kit
Super Maramu Amel
V 0.99**

Introduction

The replacement of toe pulleys is made in 4 phases :

- disassembly of current pulleys,
- cleaning of the rod, bearings and sleeves,
- screwing of the new pulley on the rod, and
- setting up of the rod with pulley.



Old toe pulley

Disassembly of a current pulley

The first step of the replacement is the disassembly of old pulleys with care about the stainless steel rod (axis of rotation).

Disassembly of the rod and pulley assembly on the boat :

- loosen the rope and undo the knot of the rope,
- remove the safety pin at the bottom of the rod under the pulley,
- pull back the upper stop pin and at the same time raise the rod assembly just a little in order to get the bottom of the rod out of the bottom bearing,
- remove the upper plastic sleeve,
- go down the rod with pulley through the hole of the stainless upper tube and at the same time tilt the rod-pulley assembly (out of the bottom bearing).



Safety pin at rod bottom



Upper sleeve & stop pin

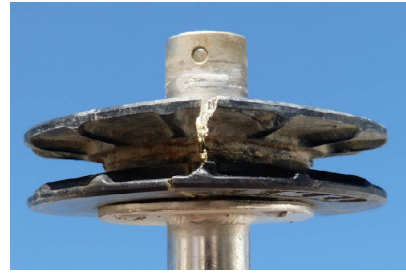
Top and bottom plastic sleeves are normally hard to remove, but pay attention not to lose or damage them.

Disassembly of the pulley

Now is the next step to pull apart the pulley from the rod. Due to the corrosion of the aluminum pulley the best way is to cut the stainless assembly screws from the middle of the groove.



Bottom view



Side view

Be careful not to cut the stainless rod !

Use an electric grinder with a thin disk for metal on an holder to do precise cuts, parallel to the top of the pulley.



Screws cuttings

Depending of the strength of the pulley and the “corrosion gluing” you can have to cut the groove almost up to the stainless rod !

Normally the pulley breaks apart when screws are cut. But the stainless rod can be misshaped and disassembly difficult.
On the test boat one pulley breaks in 6 parts as soon as the 3d screw was cut.
The other pulley was severely corroded but remained glued to the rod by the corrosion : a larger area had to be cut.
Don't hesitate at using a ball bearing puller tool, but pay attention to not bend the rod.

If the stainless rod is misshaped by the corrosion, use a plain stainless steel rod of diameter just below the rod hole diameter to reshape it with a hammer before continuing.



Corrosion misshaped rod

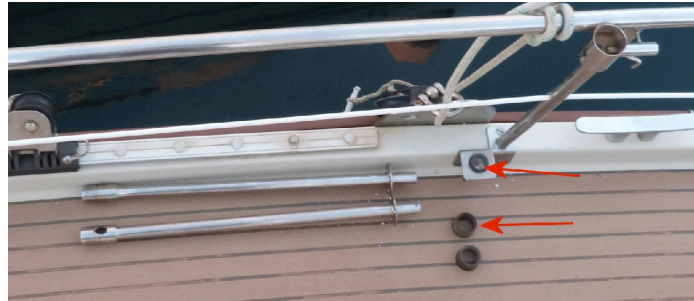
Clean carefully all pieces, paying attention to the soft plastic parts and to traces of corrosion on all parts.

The whole traces of corrosion must have **disappeared**.

Assembly of the new pulley

The assembly uses the opposite path :

- first assemble the pulley on the rod and
- next put back the rod in its bearings.

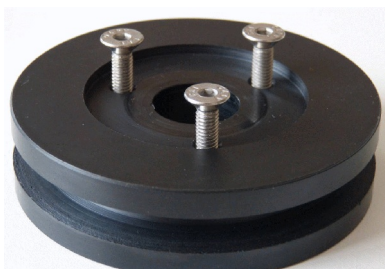


Rods ready to reassemble

Assembly of the pulley on the rod :

- put the pulley on the bottom of the stainless rod,
- put the 3 screws paying attention to put the screws perpendicular to the pulley and
- tighten the screws (**torque not more than 1 mN**).

Do not hesitate at using general purpose **threadlocker** of medium bond strength.



Screws perpendicular to pulley



Prototype assembly

Assembly of the rod on boat :

- **grease** bottom and upper sleeves before putting the rod,
- tilt the rod and put its upper part into the upper sleeve from below (with the stop pin pulled),
- raise the rod above the upper sleeve,
- lower the rod when putting the bottom end into the bottom sleeve,
- check the clearance of the rod in the sleeves and its free rotation and
- put the safety pin on the rod hole at its bottom end.



Safety pin hole



Upper bearing



Rod fitted with grease

The assembly looks like this picture :



Prototype on boat

Next reassemble the rope in the groove and make the knot on the trolley.

Enjoy.

Troubleshootings

In case of troubleshootings mails can be sent to ba@ancelle.net with details and pictures.

Content of the kit

2 toe pulleys

6 stainless steel screws